

Read csv file in R with double quotes

Asked 7 years, 9 months ago Modified 7 years, 9 months ago Viewed 27k times  Part of [R Language Collective](#)



Suppose I have a csv file looks like this:

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desired output should be:



```
df <- data.frame(Type='A',ID=3, NAME=NA, CONTENT='I have comma, ha!',
                  RESPONSE='I have open double quotes\''', GRADE=A, SOURCE=NA)
df
  Type ID NAME          CONTENT          RESPONSE GRADE SOURCE
1    A  3  NA I have comma, ha! I have open double quotes"    A    NA
```

I tried to use `read.csv`, since the data provider uses quote to escape comma in the string, but they forgot to escape double quotes in string with no comma, so no matter whether I disable quote in `read.csv` I won't get desired output.

How can I do this in R? Other package solutions are also welcome.

[r](#) [csv](#)

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asked Aug 19, 2015 at 19:04



[Bamqf](#)

3,342 8 33 47

`read.csv("data.csv", quote='', sep=',') doesn't work because the content column will be wrong.`
`read.csv("data.csv", quote='') doesn't work because the columns are not separated properly – nathanesau`
Aug 19, 2015 at 19:59

3 Answers

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`fread` from `data.table` handles this just fine:

10



```
library(data.table)

fread('Type,ID,NAME,CONTENT,RESPONSE,GRADE,SOURCE
A,3,"","I have comma, ha!","I have open double quotes",A,""')
#   Type ID NAME          CONTENT          RESPONSE GRADE SOURCE
#1:    A  3      I have comma, ha! I have open double quotes"    A
```



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answered Aug 19, 2015 at 20:10



[eddi](#)

49k 6 102 155

Nice one. Should have thought of that myself. – [David Arenburg](#) Aug 19, 2015 at 21:29

▲ I'm not too sure about the structure of CSV files, but you said the author had escaped the comma in the text under content.

2

▼ This works to read the text as is with the " at the end.



```
read.csv2("Test.csv", header = T, sep = ",", quote="")
```



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answered Aug 19, 2015 at 19:28



[Buzz Lightyear](#)

814 1 7 18

1 How is this different from just `read.csv(text = text, quote="")` ? Though this also doesn't give the desired output I guess. – [David Arenburg](#) Aug 19, 2015 at 19:31

Tried this solution. Columns get shifted so it won't get desired output. – [Bamqf](#) Aug 19, 2015 at 19:35

Actually with this solution the column are also being split. – [David Arenburg](#) Aug 19, 2015 at 19:39

This is not valid CSV, so you'll have to do your own parsing. But, assuming the convention is as follows, you can just toggle with `scan` to take advantage of most of its abilities:

2

1. If the field starts with a quote, it is quoted.
2. If the field does not start with a quote, it is raw

```
next_field<-function(stream) {  
  p<-seek(stream)  
  d<-readChar(stream,1)  
  seek(stream,p)  
  if(d=="\"")  
    field<-scan(stream,"",1,sep=",",quote="\"",blank=FALSE)  
  else  
    field<-scan(stream,"",1,sep=",",quote="",blank=FALSE)  
  return(field)  
}
```

Assuming the above convention, this sufficient to parse as follows

```
s<-file("example.csv",open="rt")  
header<-readLines(s,1)  
header<-scan(what="",text=header,sep=",")  
line<-replicate(length(header),next_field(s))  
  
setNames(as.data.frame(lapply(line,type.convert)),header)
```

	Type	ID	NAME	CONTENT	RESPONSE	GRADE	SOURCE
1	A	3	NA	I have comma, ha! I have open double quotes"		A	NA

However, in practice you might want to first write back the fields, quoting each, to another file, so you can just `read.csv` on the corrected format.

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answered Aug 19, 2015 at 21:28



A. Webb

26.1k 1 63 95